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Editors
Nano Letters
American Chemical Society

Dear Editors:

Please consider our Letter, "Reversible β-Sheet-like Ordering and XPCS-Resolved Dynamic Arrest during Elastin-like Polypeptide Phase Separation," for publication in *Nano Letters*.

We report a sequence-defined elastin-like polypeptide that forms a hierarchical protein-rich assembly containing thermally reversible β-sheet-like local order. Temperature-dependent SAXS/WAXS resolves the reversible molecular-to-network structural transition, while isothermal small-angle X-ray photon correlation spectroscopy shows a two-step decay in which the fraction of slow mesoscale dynamics progressively increases across 76–167 nm. The resulting physical picture is unusual: amyloid-like local packing remains thermally labile even as the assembled network becomes dynamically arrested on the experimental time window.

The principal advance is the direct separation of structural assembly from dynamical arrest across molecular and mesoscale length scales. High-speed SA-XPCS resolves the progressive loss of the first decay and growth of a slow component, behavior that is not evident from the time-averaged scattering profile alone. The interpretation is deliberately bounded by the measurements: the high-q ordering and low-q dynamics were obtained in separate experiments, so the ordered contacts are identified as candidate physical junctions rather than asserted to be the uniquely proven cause of arrest.

This work fits the scope of *Nano Letters* at the intersection of sequence-defined nanomaterials, protein self-assembly, nonequilibrium soft matter, and coherent X-ray characterization. It provides a nanoscale structure–dynamics framework for designing responsive protein materials in which β-sheet-like local order can be thermally switched without permanent aggregation. Rapid publication is timely because high-speed coherent X-ray methods are opening direct access to nonequilibrium dynamics in biomolecular nanomaterials.

The manuscript is original, is not under consideration elsewhere, and has been approved by all authors. The authors declare no competing financial interest. Correspondence may be addressed to Qingteng Zhang (qzhang234@anl.gov), Arvind Ramanathan (ramanathana@anl.gov), or Gyorgy Babnigg (gbabnigg@anl.gov).

Thank you for your consideration.

Sincerely,

Qingteng Zhang, Ph.D.
on behalf of all authors

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Suggested Reviewers

The following researchers have relevant expertise in elastin-like polypeptides, protein-based materials, arrested phase separation, and X-ray photon correlation spectroscopy.

Bradley D. Olsen
Alexander and I. Michael Kasser (1960) Professor of Chemical Engineering, Massachusetts Institute of Technology
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Expertise: elastin-like polypeptide gels, associative protein polymers, and arrested phase separation.

Ashutosh Chilkoti
Alan L. Kaganov Distinguished Professor of Biomedical Engineering, Duke University
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Expertise: recombinant elastin-like polypeptides, thermoresponsive protein materials, and hierarchical self-assembly.

Fajun Zhang
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Expertise: protein phase separation, SAXS, and XPCS of protein-rich systems.

Christian Gutt
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Expertise: XPCS, nonequilibrium dynamics, and dense protein systems.

Francesco Sciortino
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Expertise: colloidal gelation, arrested phase separation, and network-forming fluids.